

8 - Constraint Satisfaction Problems

Artificial Intelligence Lab Manual - Clean Study Edition

Learning Objectives

- Model a problem using variables, domains, and constraints.
- Implement recursive backtracking search.
- Use consistency checks and variable-ordering heuristics to reduce search.

Background

A CSP separates problem structure from the search procedure. Map coloring, Sudoku, timetabling, and N-Queens can all be expressed using variables, domains, and constraints. Early detection of inconsistency prevents wasted exploration.

Core Concepts

Concept	Meaning in this lab
Variable	Unknown whose value must be assigned.
Domain	Allowed values for a variable.
Constraint	Rule restricting combinations of assignments.
Backtracking	Depth-first construction of a consistent assignment.
MRV	Choose the unassigned variable with the fewest legal values first.

Guided Procedure

1. Model map coloring with one variable per region and a small color domain.
2. Write a consistency function that rejects equal colors on neighboring regions.
3. Implement recursive backtracking.
4. Add the minimum-remaining-values heuristic for variable selection.
5. Compare number of recursive calls with and without the heuristic.

Reference Implementation

Use the following as a starting point. Read and modify it rather than treating it as a final answer.

```
def backtrack(variables, domains, neighbors, assignment):
    if len(assignment) == len(variables):
        return assignment

    var = min(
        (v for v in variables if v not in assignment),
        key=lambda v: len(domains[v])
    )
    for value in domains[var]:
        ok = all(assignment.get(n) != value for n in neighbors[var])
        if ok:
            assignment[var] = value
            result = backtrack(variables, domains, neighbors, assignment)
            if result:
                return result
            del assignment[var]
    return None
```

Lab Exercises

Task 1. Solve a small map-coloring problem with three colors.

Task 2. Implement N-Queens as a CSP.

Task 3. Add forward checking by temporarily removing inconsistent values from neighboring domains.

Task 4. Explain how AC-3 differs from plain backtracking consistency checks.

Suggested Deliverables

- Notebook or Python file containing the completed implementation.
- Short result table or output evidence for each required comparison.
- A concise interpretation of what changed and why; do not submit output without explanation.

Review Questions

1. What are the three main components of a CSP?
2. Why can MRV reduce branching?
3. What information is propagated by forward checking?

Completion check: You should be able to explain the algorithm or workflow in your own words, reproduce the main result, and identify at least one limitation.